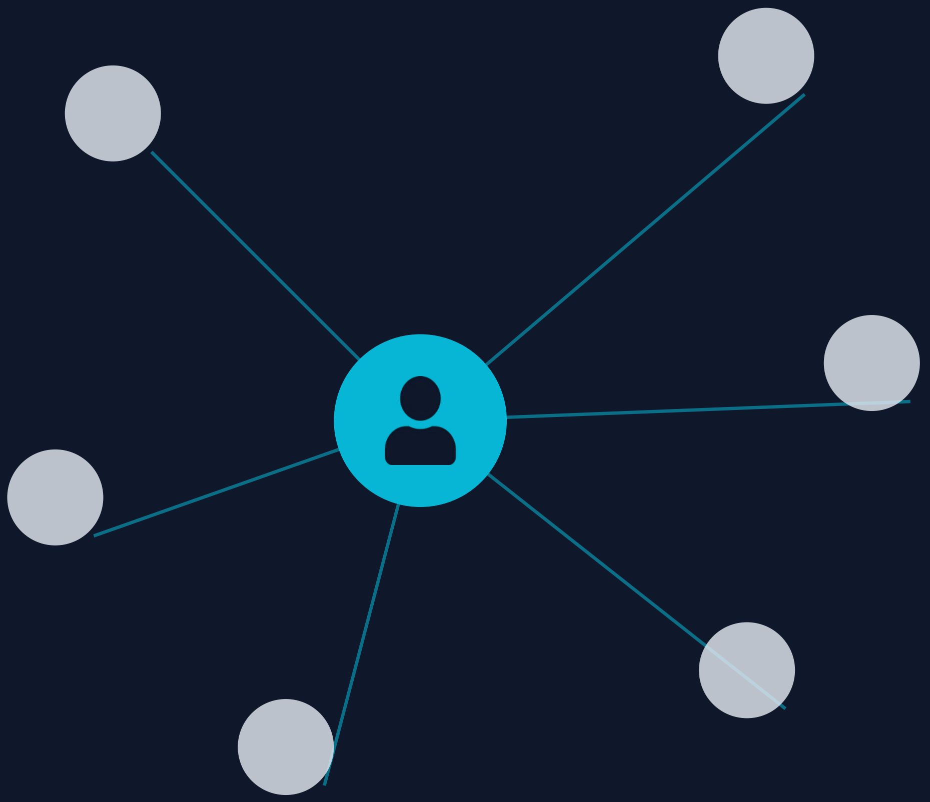
E-COMMERCE · ARCHITECTURE & IMPLEMENTATION

Recommendation Systems: Architecture & Implementation

Retrieval, ranking, serving, and evaluation at scale

A technical deep-dive for the engineering team



Presented by Your Name · June 2026

PROBLEM FRAMING

Top-K over f(user, item, context) under tight constraints

Goal: rank the catalog by predicted relevance and return the top-K, where `score = f(user, item, context)`.

Latency

Score & rank within a page-load budget (tens of ms).

Scale

Catalogs of 10^6–10^8 items; millions of QPS at peak.

Freshness

New items, new behavior, and context must propagate fast.

Objective

Optimize a business metric, not just offline accuracy.

REFERENCE ARCHITECTURE

A multi-stage funnel with separate serving & training planes

Request
user + context

→

Candidate gen
ANN retrieval $\sim 10^3$

→

Ranking
DNN / GBDT scorer

→

Re-ranking
rules · diversity

Top-K → UI

Serving plane (per request, real-time)

Supporting planes (mostly batch)

Feature store

Online + offline features with point-in-time correctness; serves both training and inference.

Embedding / ANN index

Item & user vectors built in batch, loaded into an ANN index (HNSW / IVF-PQ) for retrieval.

Training pipeline

Scheduled retraining, offline eval, model registry, and gated rollout to serving.

STAGE 1 — CANDIDATE GENERATION

High-recall retrieval, cheap per item

Reduce 10^6–10^8 items to ~10^3 candidates in milliseconds. Optimize recall; precision is the ranker's job.

Two-tower retrieval

Separate user & item encoders; dot-product in shared space; ANN at serve time.

Graph embeddings

Item graph from behavior sequences (e.g., Alibaba EGES) with side info for cold start.

Co-occurrence

Market-basket / item-to-item from co-purchase & co-view logs.

Sources & blending

Multiple recall sources (recent, popular, personalized) merged and de-duped.

ANN backends: HNSW, IVF-PQ, ScaNN — tune recall vs. latency vs. memory.

STAGE 2 — RANKING MODELS

From linear models to sequence-aware deep nets

1

LR / GBDT

Hand-crafted features; strong, cheap baselines still in production today.

→

2

Wide & Deep

Memorization (wide) + generalization (deep) over sparse embeddings.

→

3

DIN / DLRM

Attention over user history; large embedding tables for sparse features.

→

4

SASRec / Transformers

Self-attention over interaction sequences for session intent.

Inputs: dense features + sparse-ID embeddings. Train on implicit feedback (clicks/purchases) with negative sampling.

Garbage in, garbage ranked

Explicit

sparse, low-noise

- Ratings, reviews
- Likes, wishlists
- Stated preferences

Implicit

abundant, biased

- Clicks, impressions
- Purchases, returns
- Dwell, scroll depth, cart events

Watch for: exposure & position bias in implicit logs (model as weighted positives, not absent negatives) · train/serve skew (shared feature store) · point-in-time correctness to prevent label leakage.

COLD START, TECHNICALLY

Bootstrap entities with no interaction vector

New user

No interaction history → CF cannot place them in latent space.

New item

No interactions → embedding untrained and unreachable by retrieval.

Content / side features

Bootstrap item vectors from attributes; EGES-style side-info aggregation.

Popularity priors

Time-decayed popularity as a fallback recall source.

Exploration (bandits)

Deliberately allocate impressions to gather signal on new items.

Onboarding & context

Seed explicit preferences; use device, geo, and session context.

EVALUATION METHODOLOGY

Offline gates; online decides

Offline (proxy)

- Precision@k, Recall@k — retrieval quality
- NDCG, MAP — ranking quality (top-weighted)
- AUC — pairwise ranking / CTR calibration
- Fast & cheap, but only a proxy for impact

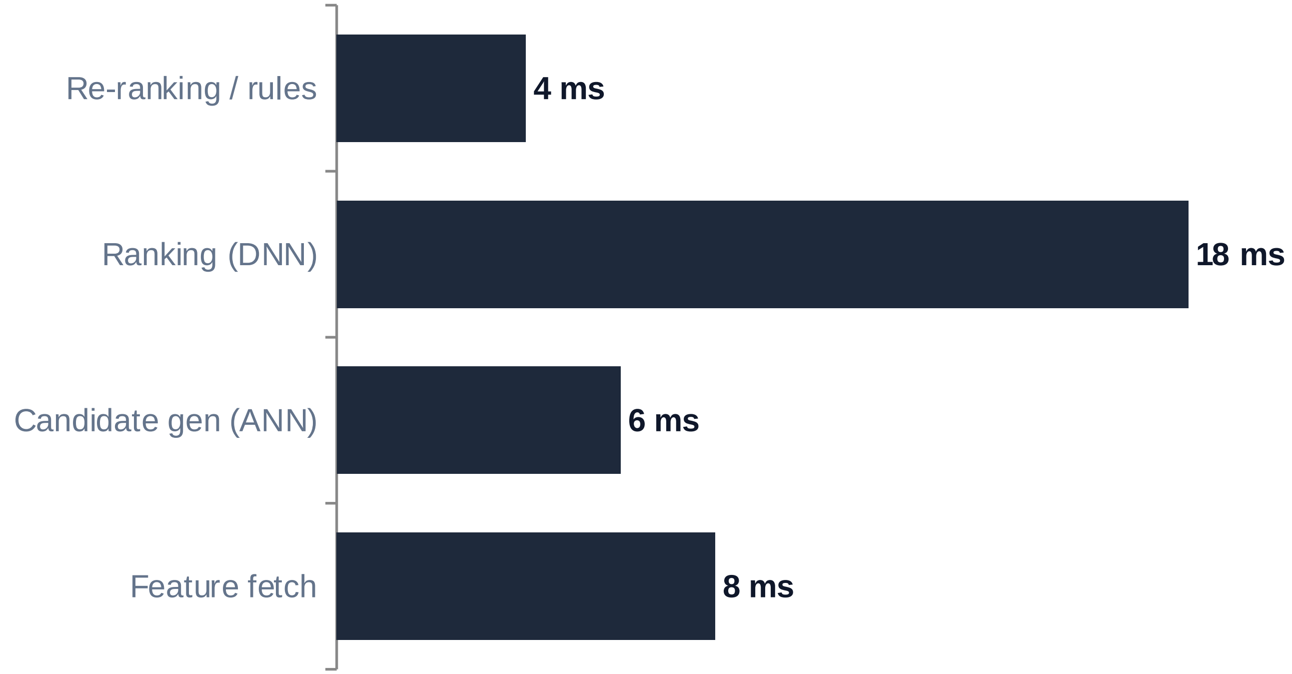
Online (ground truth)

- A/B tests with statistical significance
- Interleaving for sensitive ranking comparisons
- Primary: CTR, CVR, revenue / AOV
- Guardrails: latency, diversity, returns

Mind the offline/online gap: better NDCG ≠ better revenue. Validate every launch online.

Where the milliseconds go (and how to save them)

Illustrative per-request budget within a page-load SLA. Heavy compute is precomputed in batch.



Batch vs. real-time

Batch

Embeddings, ANN index, candidate lists, features

Real-time

Feature fetch, ranking, re-ranking, fallback

Levers

Caching, model distillation, quantization, early-exit

PRODUCTION CASE STUDIES

How leaders solved scale, sparsity & cold start

Taobao / Alibaba

EGES: build an item graph from behavior, learn embeddings, fold in side info (category, brand, price) for cold start.

A/B tested CTR gains on the Mobile Taobao home feed at billion-item scale.

JD.com (Jingdong)

Deep semantic retrieval + a pairwise deep re-ranker for product search and recommendations.

Millisecond retrieval and personalized ranking across a $100B+ GMV catalog.

Amazon

Item-to-item collaborative filtering: precompute an item-item similarity matrix offline.

O(1)-style online lookups; scales where user-based kNN does not.

CHALLENGES & MITIGATIONS

The hard problems, with concrete levers

Feedback loops

Exploration / bandits to break self-reinforcement.

Popularity bias

Inverse-propensity weighting; debiased training.

Diversity

MMR or DPP re-ranking for coverage vs. relevance.

Privacy

Consent, minimization, on-device / federated options.

Fairness

Audit exposure parity across users and sellers.

Position bias

Model click position; debias from logged impressions.

STACK & BUILD DECISIONS

A reference toolkit and the choices that matter

Layer	Options	Decision driver
Retrieval / ANN	FAISS · ScaNN · HNSWlib · vector DB	recall vs. latency vs. ops cost
Modeling	GBDT (XGBoost/LightGBM) · TensorFlow · PyTorch	team skills · model complexity
Features	Feature store (e.g. Feast) · streaming	train/serve parity · freshness
Serving	Model server · cache · autoscaling	p99 latency · cost
Experimentation	A/B platform · interleaving · metrics	trustworthy decisions

Build vs. buy: buy retrieval/feature infra to start; build the ranker where domain data is your edge. Roll out behind flags with A/B gating.

KEY TAKEAWAYS

Engineering takeaways

1. Recommendation = constrained top-K over f(user, item, context)
2. Retrieval → ranking → re-ranking, with batch/real-time decoupling
3. Embeddings + ANN for recall; GBDT/DNN/sequence models for ranking
4. Treat implicit feedback as biased signal; fix train/serve skew
5. Offline metrics gate; online A/B decides — design for the gap

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Let's align on the reference architecture and a build plan.

Thank you

APPENDIX

Production systems deep-dive

R&D history (pre- vs post-GPT) and per-segment effectiveness for Amazon, Taobao / Alibaba, and JD.com. All quantitative figures from 2024–2026 publications.

Amazon — recommendation system history

From item-to-item CF to a RAG-based generative assistant + commonsense knowledge graph.

PRE-GPT ERA			GPT / LLM ERA	
1998	**2003**	**2017**	**2024**	**2025**
Item-to-item CF	**Published (IEEE)**	**Whole-store personalization**	**Rufus + COSMO**	**Rufus at scale**
Offline item–item similarity matrix; O(1) online lookup.	Linden et al., IEEE IC; scales since items ≪ users.	Deep models; per-widget personalization, A/B-tested.	Custom shopping LLM on Bedrock; RAG + COSMO knowledge graph.	Account-level memory across Kindle/Prime/Audible; per-user conditioning.

GPT-era impact

~$12B	300M+	+60%
incremental annualized sales (Q4'25)	users · 149% YoY MAU	purchase-completion (engaged cohort)

Sources: Amazon Science (2024); Amazon Q4 2025 earnings; COSMO — ACM SIGMOD 2024; Linden et al., IEEE Internet Computing (2003, historical).

Taobao / Alibaba — recommendation system history

From graph embeddings/DIN to LLM-based generative retrieval and ranking.

PRE-GPT ERA

2018

EGES graph embeddings

Graph embeddings + side info; powers “Guess What You Like.”

2018

Deep Interest Network

Attention over behaviour history for CTR (KDD 2018).

GPT / LLM ERA

2025

LUM — Large User Model

Scaling-law user model; +CTR/RPM in Taobao search A/B.

2025

URM universal retriever

LLM as universal retriever; tens-of-ms latency.

2025

RecGPT

Intent-centric LLM framework; reasoning over log-fitting.

GPT-era impact

+2.9% / +1.2%

CTR / RPM · Taobao search A/B — LUM (2025)

18.8% → 46.2%

new-category share · serendipity retrieval — URM (2025)

Sources: Wang et al., KDD 2018 (EGES); Zhou et al., KDD 2018 (DIN); Yan et al., 2025 (LUM); Jiang et al., 2025 (URM, arXiv:2502.03041); Yi et al., 2025 (RecGPT).

JD.com (Jingdong) — recommendation system history

From deep retrieval to generative CTR and generative retrieval in production.

PRE-GPT ERA		GPT / LLM ERA		
2017	2021	2025	2025	2025
Telepath	**Semantic retrieval + re-rank**	**GenCTR**	**GRAM**	**CFRAG**
Deep-learning, vision-aware recommender (2017).	Deep semantic retrieval + pairwise re-ranking (2021).	Generative pretraining → discriminative CTR transfer.	Generative retrieval & alignment for e-commerce.	LLM-embedding-augmented recall (RAG for rec).

GPT-era impact

Gen → Disc

GenCTR transfer in production (2025)

Gen retrieval

GRAM & CFRAG for JD search (2025)

Sources: Zhao et al. (Telepath), 2017; Li et al., 2021; Kong et al., 2025 (GenCTR, arXiv:2507.11246); JD GRAM & CFRAG, 2025.

Effectiveness by user segment (LLM era)

Empirically, generative/LLM methods yield the largest lift on cold-start segments; power users benefit from long-sequence user modeling.

Cold-start (new user/item)

Sparse or no interactions

- **Mechanism:** content/semantic features + exploration bootstrap missing embeddings
- **URM (Alibaba):** serendipity retrieval: new-category share 18.8%→46.2%
- **Cold-start studies (2025):** +120% Recall@200 coldest items; Netflix +8% novel recall

Casual / mid-tail

Short, intermittent sequences

- **Intent reasoning:** RecGPT replaces log-fitting with intent inference
- **Conversational RAG:** Rufus conditions on query + retrieved evidence
- **Semantic IDs:** generalize across sparse behaviour

Head / power users

Long, dense sequences

- **Scaling laws:** LUM: long-sequence user model → +2.9% CTR / +1.2% RPM
- **Engaged cohort:** Amazon Rufus +60% purchase completion
- **Sequence transducers:** HSTU/SASRec-style attention over deep history

Sources (2025): Jiang et al. (URM); Yan et al. (LUM); Amazon Q4 2025; Li et al., Nov 2025 (Netflix); cold-start video study, Aug 2025. Platform-level per-segment figures are rarely disclosed; cross-industry studies used as directional evidence.

APPENDIX · SOURCES

LLM-era sources (2024–2026)

Amazon — Rufus	Amazon Science, “The technology behind Amazon's GenAI shopping assistant, Rufus” (2024); scale & sales via Amazon Q4 2025 earnings.	*amazon.science/blog/…-rufus*
Amazon — COSMO	Commonsense knowledge graph built with LLMs, ACM SIGMOD 2024.	*amazon.science (COSMO, SIGMOD 2024)*
Alibaba — URM	Jiang et al., “LLM as Universal Retriever in Industrial-Scale Recommender System” (2025).	*arxiv.org/abs/2502.03041*
Alibaba — LUM & RecGPT	Yan et al., “Large User Model” (2025); Yi et al., “RecGPT Technical Report” (2025).	*arXiv 2025 (LUM); RecGPT report, Aug 2025*
JD.com — GenCTR & GRAM	Kong et al., “Generative CTR Prediction…” (2025); JD “GRAM: Generative Retrieval and Alignment” (2025).	*arxiv.org/abs/2507.11246*
User-class / cold-start	Netflix cold-start (Li et al., Nov 2025); video-platform cold-start study (Aug 2025); cold-start LLM survey (2025).	*arXiv 2025*

Note: pre-GPT milestones (item-to-item CF 1998/2003; EGES & DIN 2018; JD Telepath 2017) are historical; all quantitative effectiveness figures above are from 2024–2026 publications.